

Convergence to the Beveridge Curve

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Law of motion of unemployment rate u :

$$\dot{u}(t) = -[\lambda + f] u(t) + \lambda$$

job-finding rate

job-separation rate

Critical point: $\bar{u}$ such that $\dot{u} = 0$

$$\dot{u} = 0 \Rightarrow -(\lambda + f) \bar{u} + \lambda = 0$$

$$\Rightarrow \bar{u} = \frac{\lambda}{\lambda + f}$$

$$\Rightarrow u(\theta) = \frac{\lambda}{\lambda + f(\theta)}$$

Beveridge curve.

$$\lambda + f(\theta) = \frac{\lambda}{u}$$

$$f(\theta) = \frac{\lambda - \lambda u}{u}$$

$$\mu \theta^{1-\eta} = \frac{\lambda - \lambda u}{u}$$

$$\theta = \left[\frac{\lambda - \lambda u}{\mu \cdot u} \right]^{\frac{1}{1-\eta}} = \frac{u}{\mu}$$

$$v = \left[\frac{\lambda - \lambda u}{\mu u} \cdot u^{1-\eta} \right]^{\frac{1}{1-\eta}}$$

$$v = \left[\frac{\lambda - \lambda u}{\mu u^{\eta}} \right]^{\frac{1}{1-\eta}}$$

Beveridge curve, $v(u)$ w/ $\frac{dv}{du} < 0$

Convergence of $u(t)$ to $u(\phi)$

Solution, $\dot{u}(t) + (\lambda + \rho) u(t) = \lambda$ initial gap shrinking

$$[u(t) - u(\phi)] = [u(0) - u(\phi)] e^{-(\lambda + \rho)t}$$

$$\Rightarrow u(t) = u(\phi) + [u(0) - u(\phi)] e^{-(\lambda + \rho)t}$$

$$\dot{u}(t) = -(\lambda + \rho) [u(0) - u(\phi)] e^{-(\lambda + \rho)t}$$

$$\dot{u}(t) = -(\lambda + \rho) \times [u(t) - u(\phi)]$$

$$\dot{u}(t) + (\lambda + \rho) u(t) = (\lambda + \rho) u(\phi) = \lambda$$

↑ differential equation is satisfied

+ satisfy initial condition at $t=0$

Interpretation: Gap b/w $u(t)$ & Beveridge curve shrinks at rate $(\lambda + \rho)$

$\Rightarrow \lambda + \rho$ is speed at which unemployment converges to Beveridge curve.

Im vs :

- $\lambda \approx 3\%$ per month
- $\rho \approx 59\%$ per month
- $\Rightarrow \lambda + \rho \approx 62\%$ per month

Half time time it takes for unemployment rate to cover $1/2$ distance to Beveridge curve.

$$\frac{\ln(2)}{\lambda + \rho} = \frac{0.69}{0.62} \approx 1.2 \text{ month.}$$

Im ~ quarter, $(1/2)^3 = 1/8$ of initial distance w/ Beveridge curve is left.

$\Rightarrow$ Convergence to Beveridge curve is very fast b/c λ, ρ are large

$\Rightarrow$ Unemployment rate always on Beveridge curve

$$u = u(\theta) = \frac{\lambda}{\lambda + \rho}$$